

BIOSTAT 718

Homework Assignment 03

Due: Sunday February 16, 2025

Suggested Reading:

- Diggle, Heagerty, Zeger and Liang: Chapter 2 (Sections 2.1-2.3) and Chapter 3 (Sections 3.1-3.6)
- Fitzmaurice, Laird and Ware: Chapter 3 (Sections 3.1-3.5)

In this assignment, you will explore between and within-cluster effects in a longitudinal setting with a binary outcome. You will explore concepts parallel to those discussed in Lecture 6 for a clustered data linear regression setting. Two hundred seventy-five children were followed for up to six quarters and observed for respiratory infection and indication of vitamin A deficiency. Gender and age were also recorded (age at baseline and at each visit). The dataset, [resp.xlsx](#), is saved in long format and is available in **Data Files** on Canvas in the **Files** link in the **Resources** folder. The variables are:

id:	child ID
resp:	indicator of respiratory infection (1=infected, 0=not infected)
xp:	indicator of xerophthalmia – vitamin A deficiency (1=yes, 0=no)
sex:	0=male, 1=female
time:	visit number (1 to 6)
baseage:	baseline age in months (centered at 32 months)
season:	1-4
age:	age in months (centered at 32)

- (a) Which variables are between-cluster variables, and which are within-cluster variables? Provide descriptive statistics to defend your claims.
- (b) Fit a logistic regression model with resp as the outcome variable and predictors age, xp and sex. Carefully interpret the age effect.
- (c) Expand the previous model to include between-cluster and within-cluster age effects. Please specify your model and indicate how you constructed your predictors for age. You do not need to fit the new model in part (c), just specify your model. (Hint: see Lecture 6 slides.)
- (d) Now fit a logistic regression model with the cluster-averaged age, change in age, xp and sex. Give careful interpretations of the two age effects. Contrast them to those in part (b).
- (e) Re-fit the models in (b) and (d), but this time include season effects (treat it as a categorical predictor). Give explanations for any large differences in the age effects you observe when season is added to the models compared to your previous results. Use some descriptive statistics or graphics to support your explanations.

- (f) For the model fitted in part (e) that decomposes age into within and between cluster variables, re-fit the model accounting for clustering. Fit a marginal (GEE) model to the data.
- a. Do the interpretations of the regression coefficients differ, comparing the model fit in part (f) to the model fit in part (e)? If so, how?
 - b. With respect to statistical inference on the associations between the odds of respiratory infection and age, did the correlated data impact the results in other ways?

Please provide thorough answers (complete sentences and explanations) to all the questions above.

This assignment is due by 11:59 pm on Sunday, February 16, 2026. Please upload your assignment to our Canvas website. Please save your work in the MS Word format (so that it can be edited).